

Arulprashath Rajarajan

Backend Software Engineer — Microservices • APIs • Distributed Systems

+1 (774) 418-9238 | Arajarajan@clarku.edu | [LinkedIn](#) | [Github](#)

EDUCATION

Clark University - School of Professional Studies

Master's in Computer Science —: Software Engineering, Distributed Systems, Agile Methodologies

Aug 2024 – May 2026

Worcester, MA

Paavai Engineering College

Bachelor's in Computer Science and Engineering

Aug 2019 – May 2023

Namakkal, India

EXPERIENCE

Alchemy Techsol India Pvt Ltd (Client: Capgemini Invent)

Sep 2023 – Mar 2024

Software Developer

India

- Built and deployed scalable **backend microservices** using **Java, Spring Boot, and REST APIs**, improving response time by **22%**.
- Developed **Python backend utilities** supporting internal data workflows & automation across engineering platforms.
- Optimized **SQL queries** for high-volume transactional systems, reducing query execution time by **30%** under production load.
- Collaborated in an **Agile development environment**, delivering features across multiple production sprints using **Git-based workflows**.
- Worked with cross-functional teams to design backend solutions and translate technical implementations into product-ready features.

SOFTWARE PROJECTS

Zero Trust AI Dev Architect — *Python, FastAPI, Docker, OpenAI, JWT*

- Engineered a secure **AI gateway** implementing **zero-trust architecture** to control and monitor developer interactions with large language models.
- Designed backend services using **FastAPI** that proxy AI requests through authentication, policy enforcement, and code-sanitization layers.
- Implemented **role-based access control (RBAC)** & audit logging to track AI usage across development workflows.
- Developed a policy engine capable of detecting and blocking prompts containing **sensitive tokens, credentials, or production code**.
- Containerized services using **Docker** to simulate secure AI infrastructure for enterprise development environments.

AI Meeting Summarizer and Q&A Bot (Capstone Project) — *Python, FastAPI, OpenAI [Live](#)*

- Designed scalable **backend services** using FastAPI to process meeting transcripts and expose REST APIs for real-time query retrieval.
- Implemented **Retrieval-Augmented Generation (RAG)** pipelines enabling context-aware semantic search across historical meeting transcripts.
- Architected a modular backend system emphasizing **service separation, scalability, and maintainability**.
- Developed structured AI outputs and action-item extraction pipelines to improve usability for non-technical users.

DevSensei – AI-Powered Code Reviewer — *Python, Streamlit, OpenAI GPT-4 [Live](#)*

- Engineered an **AI-powered code analysis platform** that evaluates repository-level source code across multiple programming languages.
- Designed backend workflows to analyze code structure, readability, and performance and generate automated refactoring suggestions.
- Implemented repository-level and file-wise analysis supporting **Python, Java, JavaScript, and C++**.
- Integrated **Mermaid.js** visualizations to automatically generate code logic flowcharts for improved developer comprehension.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, SQL

Backend Engineering: Microservices Architecture, REST APIs, FastAPI, Spring Boot

Testing & Automation: Pytest, Jest, API Testing, CI/CD Pipelines

Cloud & DevOps: Docker, Jenkins, Google Cloud Platform

Systems & Networking: Linux, HTTP, TCP/IP, Distributed Systems

Tools: Git, GitHub